

Problem D. Divide and Conquer

Time Limit: 2 seconds | Memory Limit: 512 MB

Standard Input / Standard Output

Segment tree is a data structure based on the divide-and-conquer algorithm, but this has nothing to do with this problem---it's just to make the title start with D.

For this problem, a segment tree is a perfect binary tree. A segment tree of depth n has exactly $2^{n+1} - 1$ nodes. Note that the root has depth 0, so a segment tree of depth n has $n + 1$ levels.

To make the structure of a segment tree easier to describe, we assign indices to its nodes. The root is assigned index 1. Every non-leaf node is assigned an index k satisfying $1 \leq k < 2^n$. It has exactly two children: its left child has index $2k$, and its right child has index $2k + 1$.

Each node of a segment tree also has a value. Let a_k denote the value of node k . For every non-leaf node k , its value a_k must be the maximum of the values of its two children; that is,

$$a_k = \max\{a_{2k}, a_{2k+1}\}.$$

The leaf nodes have indices $2^n, 2^n + 1, \dots, 2^{n+1} - 1$, and their values must form a permutation of $1, 2, \dots, 2^n$. In other words, the elements of the sequence $a_{2^n}, a_{2^n+1}, \dots, a_{2^{n+1}-1}$ must be pairwise distinct integers, and $1 \leq a_{2^n+i} \leq 2^n$ must hold for every $0 \leq i < 2^n$.

There are now q constraints. The i -th constraint is of the form (u_i, x_i) . Here, u_i is a node index and x_i is a specified value; the constraint requires the value a_{u_i} of node u_i to be equal to x_i . Calculate the number of segment trees satisfying all q constraints, or equivalently, the number of distinct valid sequences $a_1, a_2, \dots, a_{2^{n+1}-1}$. Output the answer modulo 998244353.

Input

The first line contains two integers n ($1 \leq n \leq 18$) and q ($0 \leq q \leq 2^{n+1} - 1$).

Each of the next q lines contains two integers u_i ($1 \leq u_i < 2^{n+1}$) and x_i ($1 \leq x_i \leq 2^n$), meaning that the value a_{u_i} of node u_i is required to be equal to x_i .

Output

Output one integer: the number of valid leaf permutations modulo 998244353.

Examples

Sample Input 1	Sample Output 1
2 1 2 2	4

Sample Input 2	Sample Output 2
2 2 1 4 2 4	12

Sample Input 3	Sample Output 3
2 2 2 3 4 4	0

Sample Input 4	Sample Output 4
4 2 1 2 1 3	0